# The Number of Terms in Greedy Fibonacci–Lucas Representations

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**ABSTRACT**
We prove Kimberling's conjecture on greedy representations by Fibonacci and Lucas numbers: the ratio of the number of Zeckendorf summands of $1, \dots, n$ to the number of terms of their greedy representations tends to a limit in $(1.2, 1.4)$, and the limit is $(5-\sqrt{5})/2$. We also prove the conjectures of Greathouse and Kurkov on the number of terms of the greedy representation.

**KEYWORDS**
Fibonacci numbers; Lucas numbers; Zeckendorf representation; greedy algorithm; summatory function; formal verification

## 1. Introduction

Let $F_k$ and $L_k$ be the Fibonacci and Lucas numbers, $F_0 = 0$, $F_1 = 1$, $L_0 = 2$, $L_1 = 1$, let $\varphi = (1+\sqrt{5})/2$ and $\psi = (1-\sqrt{5})/2$, and let

$$S = \{1, 2, 3, 4, 5, 7, 8, 11, 13, 18, 21, 29, \dots\}$$

be the set of positive Fibonacci and Lucas numbers (A116470 in the OEIS [8]). For $n \geq 1$ let $g(n)$ be the largest element of $S$ that is at most $n$. The greedy algorithm subtracts $g(n)$ from $n$ and repeats; the number of terms it produces is

$$a(0) = 0, \qquad a(n) = 1 + a(n - g(n)) \quad (n \geq 1),$$

which is A214973. Let $z(n)$ be the number of terms in the Zeckendorf representation of $n$ (A007895), and let

$$W(n) = \sum_{i=1}^{n} a(i), \qquad V(n) = \sum_{i=1}^{n} z(i)$$

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be A214981 and A179180. Representations by Lucas numbers go back to Carlitz, Scoville and Hoggatt [1]. Kimberling [3, Section 6] recorded the following conjecture, also stated in A214981.

**Conjecture 1.1** (Kimberling)**.** The limit of $V(n)/W(n)$ exists and lies in the interval $(1.2, 1.4)$.

Two more conjectures, from 2021, are recorded in A214973. Let A049651$(j) = (F_{3j+1} - 1)/2$.

**Conjecture 1.2** (Greathouse)**.** For every $j \geq 1$, $a(\text{A049651}(j)) = j$, and this is the first occurrence of $j$ in A214973; that is, apart from its initial term, A049651 is the sequence of record positions of A214973.

The statement that A049651$(j)$ is the least $n$ with $a(n) = j$ appeared earlier, without proof, in a 2012 comment by Kimberling in A049651.

**Conjecture 1.3** (Kurkov)**.** For every $n \geq 1$, $a(n) = \text{A329320}(\text{A048679}(n))$.

Here, if $n = F_{k_1} + \cdots + F_{k_s}$ is the Zeckendorf representation of $n$, with $k_1 > \cdots > k_s \geq 2$ and $k_i - k_{i+1} \geq 2$, then A003714$(n) = \sum_i 2^{k_i - 2}$, whose binary word has no two adjacent ones. Preceded by a zero, such a word is a concatenation of the words 0 and 01 in exactly one way, and A048679$(n)$ is the number whose binary word is obtained by replacing each factor 01 by 1. Finally A035263$(m)$ is the parity of the 2-adic valuation $\nu_2(2m)$, and

$$\text{A329320}(y) = \sum_{k=0}^{\lfloor \log_2 y \rfloor} \big(1 - \text{A035263}(1 + \lfloor y/2^k \rfloor)\big), \qquad \text{A329320}(0) = 0.$$

For $x \geq 1$ let $K(x)$ be the index $k \geq 2$ with $F_k \leq x < F_{k+1}$, and put $K(0) = 0$. We prove the three conjectures.

**Theorem 1.4.** *Let $T(x) = W(x-1)$ and $Z(x) = V(x-1)$ for $x \geq 1$, and let $c = (5 - \sqrt{5})/2 = 2 + \psi$. For every $x \geq 1$,*

$$\big|5T(x) - K(x)\,x\big| \leq 19x, \qquad \big|5Z(x) - c\,K(x)\,x\big| \leq 13x. \tag{1.1}$$

*Consequently, for every $n \geq 1$,*

$$\Big|W(n) - \frac{1}{5}\,n \log_\varphi n\Big| \leq 10n, \qquad \Big|V(n) - \frac{c}{5}\,n \log_\varphi n\Big| \leq 10n, \tag{1.2}$$

$$\Big|\frac{V(n)}{W(n)} - c\Big| \leq \frac{51}{K(n+1) - 19} \qquad \textit{whenever } K(n+1) > 19, \tag{1.3}$$

*and $V(n)/W(n) \to (5 - \sqrt{5})/2 = 1.38196\ldots$, which lies in $(1.2, 1.4)$. In particular Conjecture 1.1 holds.*

The formula for $V(n)$ agrees with Lekkerkerker's theorem on the average number of Zeckendorf summands [5, 4], since $c/5 = 1/(\varphi^2 + 1)$. Pihko [7] proved the analogues of that theorem for the minimal and maximal Lucas representations and the maximal

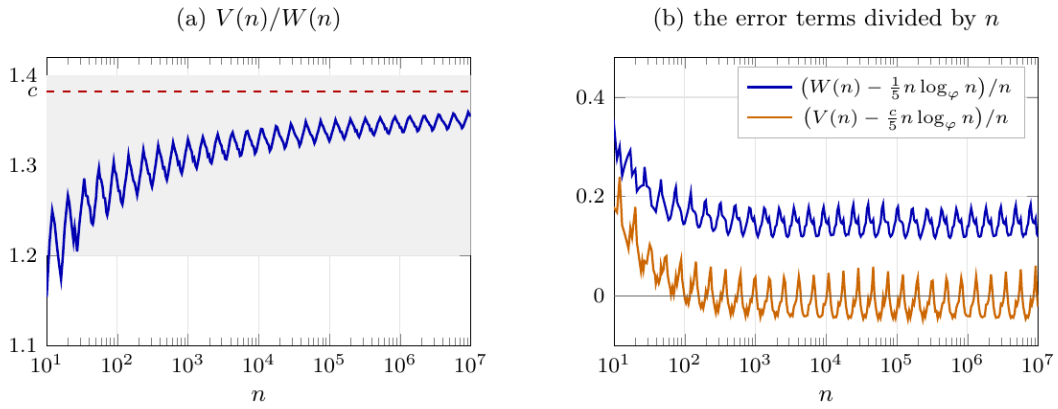



**Figure 1.** (a) The ratio $V(n)/W(n)$ for $10 \le n \le 10^7$, the conjectured interval $(1.2, 1.4)$ shaded and the limit $c = (5-\sqrt{5})/2$ dashed; the convergence is slow, since the error is of order $1/\log n$ by (1.1). (b) The two error terms of (1.2) divided by $n$; they stay far inside the bound 10.

Fibonacci representation. Theorem 1.4 shows that the greedy algorithm over $S$ uses on average $\frac{1}{5}\log_\varphi n$ terms.

**Theorem 1.5.** *For every $j \ge 1$, $a(\mathrm{A049651}(j)) = j$ and $a(m) < j$ for $0 \le m < \mathrm{A049651}(j)$. Hence Conjecture 1.2 holds.*

**Theorem 1.6.** *For every $n \ge 0$, $a(n) = \mathrm{A329320}(\mathrm{A048679}(n))$. Hence Conjecture 1.3 holds.*

The proof of Theorem 1.6 gives $a(n)$ in terms of the Zeckendorf representation of $n$: if its indices form maximal chains $k, k-2, k-4, \dots$ of lengths $\ell_1, \dots, \ell_t$, then $a(n) = \sum_i \lceil \ell_i/2 \rceil$ (Corollary 5.1). The proofs are elementary, and every statement of Theorems 1.4–1.6 is also proved in the Lean 4 proof assistant (Section 6).

## 2. The greedy step

We use $L_m = F_{m-1} + F_{m+1}$ for $m \ge 1$.

**Lemma 2.1.** *For $k \ge 4$,*

$$F_k < L_{k-1} < F_{k+1}, \qquad L_{k-1} - F_k = F_{k-2}, \qquad F_{k+1} - L_{k-1} = F_{k-3},$$

*and $S \cap [F_k, F_{k+1}) = \{F_k, L_{k-1}\}$. Hence, for $F_k \le n < F_{k+1}$,*

$$a(n) = \begin{cases} 1 + a(n - F_k), & n < L_{k-1}, \\ 1 + a(n - L_{k-1}), & n \ge L_{k-1}. \end{cases} \tag{2.1}$$

***Proof.*** The identities follow from $L_{k-1} = F_{k-2} + F_k$ and $F_{k+1} - F_k - F_{k-2} = F_{k-1} - F_{k-2} = F_{k-3}$, and $F_{k-2}, F_{k-3} \ge 1$ gives the inequalities. The only Fibonacci number in $[F_k, F_{k+1})$ is $F_k$. For the Lucas numbers, $L_0 = 2 < F_k$, the sequence $(L_m)_{m \ge 1}$ is increasing, $L_{k-2} = F_{k-3} + F_{k-1} \le F_{k-2} + F_{k-1} = F_k$, and $L_k = F_{k-1} + F_{k+1} > F_{k+1}$. So $L_{k-1}$ is the only Lucas number in $(F_k, F_{k+1})$. □

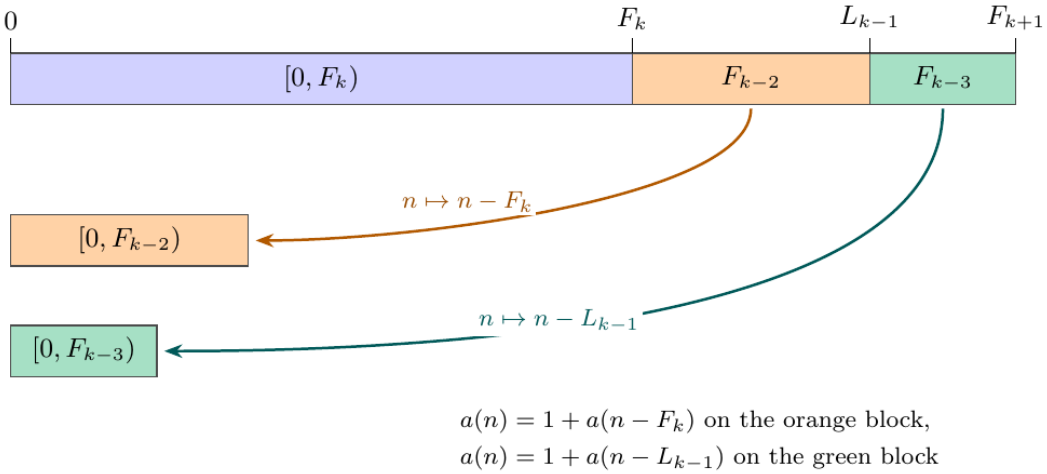



**Figure 2.** The greedy step on $[F_k, F_{k+1})$, drawn to scale for $k = 9$. By Lemma 2.1 the block $[F_k, L_{k-1})$, of length $F_{k-2}$, is moved onto $[0, F_{k-2})$ by subtracting $F_k$, and the block $[L_{k-1}, F_{k+1})$, of length $F_{k-3}$, onto $[0, F_{k-3})$ by subtracting $L_{k-1}$. The same three blocks give the decompositions (3.1) and (3.2) of the summatory function.

For $1 \le n \le 2$ we have $a(n) = 1$. The Zeckendorf representation satisfies $z(n) = 1 + z(n - F_k)$ for $F_k \le n < F_{k+1}$, $k \ge 2$, since its leading term is the largest Fibonacci number at most $n$.

**Lemma 2.2.** *Let $n = F_{k_1} + \cdots + F_{k_s}$ be the Zeckendorf representation of $n \ge 1$. If $s \ge 2$ and $k_2 = k_1 - 2$, then $a(n) = 1 + a(F_{k_3} + \cdots + F_{k_s})$; otherwise $a(n) = 1 + a(F_{k_2} + \cdots + F_{k_s})$.*

***Proof.*** If $k_1 \le 3$ then $n \in \{1, 2\}$, $s = 1$ and $a(n) = 1$. Let $k = k_1 \ge 4$, so $F_k \le n < F_{k+1}$, and let $r = n - F_k$. If $k_2 = k - 2$, then $n \ge F_k + F_{k-2} = L_{k-1}$ and $n - L_{k-1} = F_{k_3} + \cdots + F_{k_s}$. Otherwise $r = 0$ or $r < F_{k_2+1} \le F_{k-2}$, so $n < L_{k-1}$. In both cases (2.1) gives the claim. □

So the greedy algorithm over $S$ is the Zeckendorf algorithm in which a leading summand $F_k$ followed by $F_{k-2}$ is merged into $L_{k-1}$.

## 3. The summatory functions

Write $T(x) = \sum_{0 \le i < x} a(i)$ and $Z(x) = \sum_{0 \le i < x} z(i)$, so that $W(n) = T(n+1)$ and $V(n) = Z(n+1)$. If $f(i) = 1 + f(i - b)$ for $b \le i < d$, then $\sum_{i<x} f(i) = \sum_{i<b} f(i) + (x - b) + \sum_{i<x-b} f(i)$ for $b \le x \le d$. With Lemma 2.1 this gives, for $k \ge 4$,

$$T(x) = T(F_k) + (x - F_k) + T(x - F_k), \qquad F_k \le x \le L_{k-1}, \tag{3.1}$$
$$T(x) = T(L_{k-1}) + (x - L_{k-1}) + T(x - L_{k-1}), \qquad L_{k-1} \le x \le F_{k+1}, \tag{3.2}$$

and, for $k \ge 2$,

$$Z(x) = Z(F_k) + (x - F_k) + Z(x - F_k), \qquad F_k \le x \le F_{k+1}. \tag{3.3}$$

**Lemma 3.1.** *Let $e_k = 5T(F_k) - kF_k$. Then $e_{k+1} = e_k + e_{k-2} + e_{k-3}$ for $k \geq 4$, and $|e_k| \leq 2F_k$ for $k \geq 1$.*

***Proof.*** Taking $x = L_{k-1}$ in (3.1) and $x = F_{k+1}$ in (3.2),

$$T(F_{k+1}) = T(F_k) + T(F_{k-2}) + T(F_{k-3}) + F_{k-1} \qquad (k \geq 4).$$

Multiplying by 5 and subtracting $(k+1)F_{k+1}$, the recursion for $e_k$ follows from

$$kF_k + (k-2)F_{k-2} + (k-3)F_{k-3} + 5F_{k-1} = (k+1)F_{k+1},$$

which holds because $F_{k-2} + F_{k-3} = F_{k-1}$ reduces it to $5F_{k-1} - 2F_{k-2} - 3F_{k-3} = F_{k+1}$, and both sides equal $2F_{k-1} + F_{k-2}$. Since $T(1) = 0$, $T(2) = 1$ and $T(3) = 2$, we get $(e_1, e_2, e_3, e_4) = (-1, -2, -1, -2)$, so $|e_k| \leq 2F_k$ for $k \leq 4$, and by induction $|e_{k+1}| \leq 2(F_k + F_{k-2} + F_{k-3}) = 2F_{k+1}$. □

**Lemma 3.2.** *For $k \geq 1$, $5Z(F_k) = k(2F_k - F_{k-1}) - 2F_k$, and $|5Z(F_k) - c\,kF_k| \leq 4F_k$.*

***Proof.*** The formula holds for $k = 1, 2, 3$, where $Z(F_k) = 0, 0, 1$. Taking $x = F_{k+1}$ in (3.3) gives $Z(F_{k+1}) = Z(F_k) + F_{k-1} + Z(F_{k-1})$ for $k \geq 2$, and the formula for $k+1$ follows from the formulas for $k$ and $k-1$ and the identity

$$k(2F_k - F_{k-1}) + (k-1)(2F_{k-1} - F_{k-2}) + 3F_{k-1} - 2F_k = (k+1)(2F_{k+1} - F_k) - 2F_{k+1},$$

in which, after $F_{k+1} = F_k + F_{k-1}$, the coefficients of $k$ agree because $F_{k-1} + F_{k-2} = F_k$, and the remaining terms are $-(2F_{k-1} - F_{k-2}) + 3F_{k-1} - 2F_k = -F_k$ on the left and $(2F_{k+1} - F_k) - 2F_{k+1} = -F_k$ on the right. For the bound, $\psi F_k + F_{k-1} = \psi^k$ by induction on $k$ (using $\psi^2 = \psi + 1$), so

$$5Z(F_k) - c\,kF_k = -k(F_{k-1} + \psi F_k) - 2F_k = -k\psi^k - 2F_k,$$

and $|\psi| < 1$ and $k \leq 2F_k$ give $|5Z(F_k) - c\,kF_k| \leq k + 2F_k \leq 4F_k$. □

**Lemma 3.3.** *If $r \geq 0$ and $k \geq K(r)$, then $0 \leq (k - K(r))\,r \leq 2F_k$.*

***Proof.*** The claim is clear if $r = 0$ or $k = K(r)$. Otherwise let $j = K(r) < k$ and $d = k - j - 1 \geq 0$. Then $r < F_{j+1}$, and $F_k = F_jF_d + F_{j+1}F_{d+1} \geq F_{j+1}F_{d+1}$ together with $d + 1 \leq 2F_{d+1}$ gives $(k - j)\,r \leq (d+1)F_{j+1} \leq 2F_k$. □

**Proposition 3.4.** *For every $x \geq 0$, $|5T(x) - K(x)\,x| \leq 19x$.*

***Proof.*** By induction on $x$. For $x = 0, 1, 2$ the left side is 0, 2 and 1. Let $x \geq 3$ and $k = K(x) \geq 4$.

If $x \leq L_{k-1}$, let $r = x - F_k$. Then $r \leq F_{k-2}$, so $K(r) \leq k$, and by (3.1)

$$5T(x) - kx = e_k + 5r + \big(5T(r) - K(r)\,r\big) - \big(k - K(r)\big)r.$$

By Lemmas 3.1 and 3.3 and the induction hypothesis, this lies between $-4F_k - 14r$ and $2F_k + 24r$. Since $r \leq F_k$ and $x = F_k + r$, both bounds are at most $19x$ in absolute value.

If $x > L_{k-1}$, let $r = x - L_{k-1} < F_{k-3}$. By (3.2) and (3.1) at $x = L_{k-1}$, and since $x = F_k + F_{k-2} + r$,

$$5T(x) - kx = e_k + e_{k-2} + 3F_{k-2} + 5r + \big(5T(r) - K(r)\,r\big) - \big(k - K(r)\big)r,$$

which lies between $-4F_k + F_{k-2} - 14r$ and $2F_k + 5F_{k-2} + 24r$; both are at most $19x$ in absolute value. □

**Proposition 3.5.** *For every $x \geq 0$, $|5Z(x) - c\,K(x)\,x| \leq 13x$.*

***Proof.*** By induction on $x$; the case $x = 0$ is trivial. Let $x \geq 1$, $k = K(x) \geq 2$ and $r = x - F_k < F_{k-1}$, so $K(r) \leq k$ and $r \leq F_k$. By (3.3),

$$5Z(x) - c\,kx = \big(5Z(F_k) - c\,kF_k\big) + 5r + \big(5Z(r) - c\,K(r)\,r\big) - c\,\big(k - K(r)\big)r.$$

Since $0 < c \leq 2$, Lemmas 3.2 and 3.3 and the induction hypothesis place this between $-8F_k - 8r$ and $4F_k + 18r$, and both are at most $13x = 13F_k + 13r$ in absolute value. □

***Proof of Theorem 1.4.*** The bounds (1.1) are Propositions 3.4 and 3.5. Let $x = n + 1$ and $K = K(x)$. Then $W(n) = T(x)$ and $V(n) = Z(x)$, and

$$5V(n) - 5cW(n) = \big(5Z(x) - cKx\big) - c\big(5T(x) - Kx\big)$$

has absolute value at most $13x + 2 \cdot 19x = 51x$, while $5W(n) \geq (K - 19)\,x$. This gives (1.3), and $K(n+1) \to \infty$ gives the limit. Finally $6/5 < (5 - \sqrt{5})/2 < 7/5$ because $121 < 125 < 169$.

For (1.2), write $\lambda(t) = \log_\varphi t$. Since $F_{k+1} \leq \varphi^k$ and $\varphi^{k-2} \leq F_k$ for $k \geq 2$, and $F_K \leq x < F_{K+1}$, we have $\lambda(x) < K \leq \lambda(x) + 2$. Also $K \leq x + 1$, because $F_k \geq k - 1$ for every $k \geq 1$. Next,

$$x\lambda(x) - n\lambda(n) = \lambda(x) + n\,\frac{\log(1 + 1/n)}{\log \varphi},$$

and $n \log(1+1/n) \leq 1 < 3 \log \varphi$, because $\varphi^3 = 2\varphi + 1 > e$. Hence $0 \leq x\lambda(x) - n\lambda(n) \leq K + 3 \leq n + 5$. In the identity

$$5W(n) - n\lambda(n) = \big(5T(x) - Kx\big) + \big(K - \lambda(x)\big)x + \big(x\lambda(x) - n\lambda(n)\big)$$

the three terms lie in $[-19x, 19x]$, $[0, 2x]$ and $[0, n + 5]$, so $-19(n + 1) \leq 5W(n) - n\lambda(n) \leq 21(n + 1) + n + 5 \leq 48n$ for $n \geq 1$. In the same way, since $0 < c \leq 2$,

$$5V(n) - c\,n\lambda(n) = \big(5Z(x) - cKx\big) + c\big(K - \lambda(x)\big)x + c\big(x\lambda(x) - n\lambda(n)\big)$$

lies between $-13(n+1)$ and $13(n+1) + 4(n+1) + 2(n+5) \leq 46n$. Dividing by 5 gives (1.2). □

## 4. The records

Let $R_j = F_2 + F_5 + \cdots + F_{3j-1} = \sum_{i<j} F_{3i+2}$. Since $F_{3j+1} + 2F_{3j+2} = F_{3j+3} + F_{3j+2} = F_{3j+4}$, induction gives $2R_j + 1 = F_{3j+1}$, so $R_j = \mathrm{A049651}(j)$. Moreover $R_j < F_{3j}$ for

$j \geq 1$, since $2R_j < F_{3j+1} \leq 2F_{3j}$.

***Proof of Theorem 1.5.*** First, $a(R_j) = j$. This holds for $j = 0, 1$. For $j \geq 2$ let $k = 3j-1 \geq 5$. Then $R_j = F_k + R_{j-1}$ and $R_{j-1} < F_{k-2}$, so $F_k \leq R_j < L_{k-1}$, and (2.1) gives $a(R_j) = 1 + a(R_{j-1}) = j$.

Second, $a(m) \geq j$ implies $m \geq R_j$. We use induction on $m$. The cases $j \leq 1$ and $m \leq 2$ are clear, so let $j = t+2$ with $t \geq 0$, and let $m \geq 3$ and $k = K(m) \geq 4$. By (2.1) there is $r$ with $a(r) = a(m) - 1 \geq t+1$, where either $r = m - F_k < F_{k-2}$ or $r = m - L_{k-1} < F_{k-3}$. By induction $r \geq R_{t+1} \geq F_{3t+2}$, so $F_{3t+2} < F_{k-2}$, hence $3t+2 < k-2$ and $F_{3t+5} \leq F_k$. In both cases $m \geq F_k + r \geq F_{3t+5} + R_{t+1} = R_{t+2}$.

So $a(m) < j$ for $m < R_j$, and $a(R_j) = j$. Thus $R_j$ is the first occurrence of $j$, and since $R_1 < R_2 < \cdots$, the records of A214973 occur exactly at $R_1, R_2, \ldots$. □

## 5. Kurkov's formula

Let $n = F_{k_1} + \cdots + F_{k_s}$ be the Zeckendorf representation of $n \geq 1$. A *chain* is a maximal run $k_i, k_{i+1}, \ldots, k_{i+\ell-1}$ of consecutive indices with $k_p - k_{p+1} = 2$ for $i \leq p < i+\ell-1$; the indices split into chains of lengths $\ell_1, \ldots, \ell_t$, and consecutive chains are separated by gaps $k_p - k_{p+1} \geq 3$.

**Corollary 5.1.** $a(n) = \sum_{i=1}^{t} \lceil \ell_i/2 \rceil$.

***Proof.*** Apply Lemma 2.2 repeatedly. Along a chain it merges the first and second terms, the third and fourth, and so on, giving $\lceil \ell/2 \rceil$ terms; it never merges the last term of a chain with the first term of the next, since they differ by at least 3. □

**Lemma 5.2.** A048679$(n) = \sum_{i=1}^{s} 2^{h_i}$ *with* $h_i = k_i - 2 - (s-i)$. *The exponents satisfy* $h_1 > \cdots > h_s \geq 0$, *and* $h_i - h_{i+1} = 1$ *if and only if* $k_i - k_{i+1} = 2$.

***Proof.*** The binary word of A003714$(n)$, preceded by a zero, consists of the factors 01 occupying positions $k_i - 1, k_i - 2$ ($1 \leq i \leq s$; position 0 is the units digit), separated by zeros. These factors are disjoint because $k_{i+1} - 1 \leq k_i - 3$. Replacing each 01 by 1 moves the digit at position $k_i - 2$ down by one place for each of the $s - i$ factors below it, so it lands at position $h_i$. Finally $h_i - h_{i+1} = k_i - k_{i+1} - 1 \geq 1$ and $h_s = k_s - 2 \geq 0$. □

**Lemma 5.3.** *For* $y \geq 0$, A329320$(y) = \sum \lceil \ell/2 \rceil$, *the sum over the maximal blocks of consecutive ones in the binary word of* $y$, *of lengths* $\ell$.

***Proof.*** Since $\nu_2(2m) = 1 + \nu_2(m)$, we have $1 - \text{A035263}(1+u) = \nu_2(u+1) \bmod 2$, and $\nu_2(\lfloor y/2^k \rfloor + 1)$ is the number of consecutive ones of $y$ starting at position $k$ and going up. This number is 0 if the digit at position $k$ is 0, and, along a block of $\ell$ ones, it takes the values $\ell, \ell-1, \ldots, 1$, of which $\lceil \ell/2 \rceil$ are odd. The sum defining A329320$(y)$ runs over the positions $0, \ldots, \lfloor \log_2 y \rfloor$, which contain every block. □

***Proof of Theorem 1.6.*** For $n = 0$ both sides are 0. For $n \geq 1$, Lemma 5.2 shows that the blocks of ones of A048679$(n)$ correspond to the chains of the Zeckendorf representation of $n$, with the same lengths. Lemma 5.3 and Corollary 5.1 give A329320(A048679$(n)$) $= \sum_i \lceil \ell_i/2 \rceil = a(n)$. Figure 3 follows the argument for $n = 595$. □

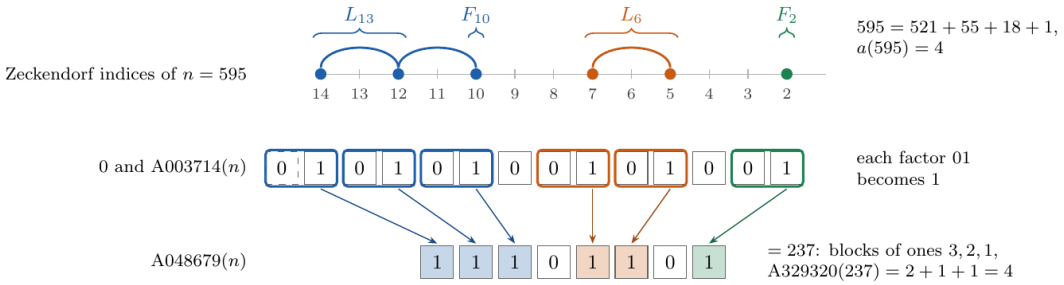



**Figure 3.** Kurkov's formula for $n = 595 = F_{14} + F_{12} + F_{10} + F_7 + F_5 + F_2$. The Zeckendorf indices form the chains $14, 12, 10$, then $7, 5$, then $2$, and the greedy algorithm uses $L_{13} = F_{14} + F_{12}$, $F_{10}$, $L_6 = F_7 + F_5$ and $F_2$, so $a(595) = 4$ (Corollary 5.1). The binary word of A003714(595), preceded by a zero, splits into the factors 01 and 0; replacing each 01 by 1 gives A048679(595) $= 237$, whose blocks of ones have the lengths of the chains (Lemma 5.2), and A329320(237) $= 4$ (Lemma 5.3).

## 6. Formal verification

Theorems 1.4, 1.5 and 1.6 are formalized in Lean 4 [2] with Mathlib [6], version 4.34.1, and the formal proofs are checked by the Lean kernel. The formalization defines the sequences directly from their entries in the OEIS: $a$ by the greedy recursion over the positive Fibonacci and Lucas numbers, $z$ as the length of the Zeckendorf representation, $W$ and $V$ as sums over $1 \le i \le n$, A048679 by the rewriting of binary digits described in Section 1, and A329320 by the displayed sum. It proves the bounds (1.1), the explicit formulas (1.2), the limit $V(n)/W(n) \to (5 - \sqrt{5})/2$ together with $6/5 < (5-\sqrt{5})/2 < 7/5$, Theorem 1.5 for every $j$, and Theorem 1.6 for every $n \ge 0$. In the formal proof of Theorem 1.6 the induction runs on the Zeckendorf representation through the recursion $\mathrm{A329320}(2^{Q+1}+u) = \mathrm{A329320}(2^{Q+1}+2^{Q}+u) = 1+\mathrm{A329320}(u)$ for $u < 2^Q$, in place of Lemma 5.3. The proofs depend only on the axioms of Lean's logic, namely propositional extensionality, choice and the soundness of quotients; they have no unproved step and add no axiom.

Independent computations in exact integer arithmetic, from the definitions of the sequences, confirm (1.1) for $x \le 10^8 + 1$, Lemma 3.1 and the formula of Lemma 3.2 for $F_k \le 10^8 + 1$, Theorems 1.5 and 1.6 and Lemma 2.2 for $n \le 10^8$, and Lemmas 2.1, 3.1 and 3.2 for $k \le 400$; they agree with the terms of the sequences recorded in the OEIS. The ratio $V(n)/W(n)$ equals 1.3421 at $n = 10^6$ and 1.3588 at $n = 10^8$.


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## Data availability

Deep Bhattacharjee computed the Lean formalization and the Julia, C, C++, shell and Python computations with the help of AI assistance (Claude, Anthropic). The Lean formalization of the proofs in this paper is archived at Zenodo, doi:10.5281/zenodo.0000000. The paper is available at `https://github.com/creelie/musical-rotary-phone/blob/main/preprintKimberlingConjecture/KimberlingConjecture.pdf`.

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